

AWS+ Devops + Docker Jenkins + Kubernetes



COURSE CURRICULUM

MODULE 1: FUNDAMENTALS OF DEVOPS

- What is DevOps?
- DevOps lifecycle and benefits
- Agile, Scrum, and ITIL integration
- Continuous Integration & Continuous Deployment (CI/CD)
- DevOps tools overview

MODULE 2: AWS DEVOPS (AMAZON WEB SERVICES)

AWS Basics

- AWS Global Infrastructure
- IAM: Identity & Access Management
- EC2, S3, VPC, CloudWatch, CloudTrail
- ELB, Route53, RDS, Lambda

AWS DevOps Tools & Services

- AWS CodeCommit (Git-based version control)
- AWS CodeBuild (Build & Test)
- AWS CodeDeploy (Automated Deployment)
- AWS CodePipeline (CI/CD Pipeline)
- AWS CloudFormation (Infrastructure as Code)
- Elastic Beanstalk

Advanced Topics

- Monitoring with CloudWatch
- Logging with CloudTrail
- Auto Scaling & Load Balancing
- AWS CLI and SDKs
- Security Best Practices

Projects

- End-to-End CI/CD Pipeline using AWS DevOps tools
- Auto-scaling web app deployment

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MODULE 3: DOCKER – CONTAINERIZATION

Docker Basics

- What is Docker?
- Docker vs Virtual Machines
- Installing Docker & Docker Architecture
- Images, Containers, Volumes, and Networks
- Docker CLI

Intermediate Concepts

- Dockerfile creation
- Building & Publishing Custom Docker Images
- Docker Compose – Multi-container applications
- Docker Hub, Private Registries
- Container Networking

Projects

- Build and containerize a Node.js/Python app
- Multi-container environment using Docker Compose

MODULE 4: JENKINS – CONTINUOUS INTEGRATION

Jenkins Basics

- Installing Jenkins (Local & AWS EC2)
- Jenkins architecture & setup
- Configuring build jobs
- Plugins & integrations

Pipeline as Code

- Freestyle Jobs vs Pipeline Jobs
- Writing Jenkinsfile (Declarative and Scripted Pipeline)
- CI/CD Pipeline with GitHub, Maven, Docker
- Notifications with Slack/Email

Projects

- CI/CD pipeline to build, test, containerize & deploy a web app using Jenkins + Docker

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MODULE 5: KUBERNETES – CONTAINER ORCHESTRATION

Introduction to Kubernetes

- Kubernetes architecture (Master, Nodes, Pods, Services)
- Minikube & kubectl
- YAML Files – Deployment, Service, Ingress

Core Components

- Pods, ReplicaSets, Deployments
- Services (ClusterIP, NodePort, LoadBalancer)
- ConfigMaps & Secrets
- Namespaces & RBAC

Advanced Concepts

- Helm Charts
- Kubernetes Networking
- Rolling updates and Rollbacks
- Auto-scaling & Health Checks
- Monitoring with Prometheus + Grafana

Projects

- Deploying a microservices-based app on Kubernetes cluster
- End-to-end pipeline with Jenkins, Docker & Kubernetes

Capstone Project (Final Week)

- Real-world project combining:
- Jenkins CI/CD
- Docker for containerization
- Kubernetes for orchestration
- AWS for hosting
- Azure DevOps for pipeline automation & cloud infrastructure
- Present project with documentation, GitHub code repo, and demo video





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